Mehmet A. Guler

Contact Information

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Education

Lehigh University, **Bethlehem, PA, US**

Ph.D. in Mechanical Engineering, January 2001

- Dissertation: "Contact Mechanics of FGM Coatings"
  Advisor: Prof. Fazil Erdogan

Lehigh University, **Bethlehem, PA, US**

M.S., Mechanical Engineering, June 1996

- Thesis: "The Problem of a Rigid Punch with Friction on a Graded Elastic Medium"
  Advisor: Prof. Fazil Erdogan

Sussex University, **Brighton, UK**

M.S., Computer Technology in Manufacturing, September 1991

- Thesis: "Self Organizing Control of Hardboard Manufacturing and Coupled Tank Systems"
  Advisor: Prof. Andrew W. Self

Middle East Technical University, **Ankara, Turkey**

B.S., Mechanical Engineering, June 1990

- Senior Design Project: Computer Aided Selection of DC Servomotors
  Advisor: Prof. I. Huseyin Filiz

Academic Experience

American University of the Middle East (AUM), **Kuwait**
Professor(Full time) **Sep, 2018 - present**
Head of Research Committee of the Mechanical Engineering Department.
Taught undergraduate courses which covered the areas of Statics, Mechanics of Materials, Machine Design, Theory of Machines and System Dynamics.

TOBB University of Economics and Technology, **Ankara, Turkey**
Professor(On leave) **Sep, 2018 - Dec, 2022**
Professor(Full time) **Oct, 2015 - Sep, 2018**
Associate Professor **Jul, 2010 - Oct, 2015**
Assistant Professor **Sep, 2006 - Jul, 2010**
Taught courses from sophomore level to graduate level, which covered the areas of statics, dynamics, machine design, finite element method for mechanical engineers and theory of elasticity.

Near East University, **Nicosia, N. Cyprus**
Professor (Part time) **Sep, 2014 - Sep, 2018**
Taught junior level machine design courses

University of Arizona, **Tucson, AZ, US**
Visiting Research Scholar **Feb, 2014 - Jun, 2014**
Conducted research in modeling hyperelastic materials using peridynamic theory for the project funded by Department of Aerospace and Mechanical Engineering of University of Arizona

Fulbright Research Scholar **Aug, 2013 - Feb, 2014**
Conducted research in modeling composite materials using peridynamic theory for the project titled "Development of a method to predict strength and failure of layered composites using peridynamic theory" funded by Fulbright

Virginia Commonwealth University, **Richmond, VA, US**
Visiting Research Scholar **Jun, 2007 - Sep, 2007**
Conducted research in springback prediction of Advanced High Strength Steels used in automobile components.

Lehigh University, **Bethlehem, PA, US**
Research Assistant **Jan, 1994 - Sep, 2000**
Research focused on stress analysis and fracture characterization of ceramic coatings on metal substrates, involving both mechanical and thermal stresses. Developed analytical models to study the contact mechanics of Functionally Gradient Materials (FGMs).

Courses Offered

TOBB University of Economics and Technology (TOBB ETU), **Ankara, Turkey**
MAK 104 Statics [(Spring – 2007, 2008, 2017) (Summer – 2008)]
MAK 203 Dynamics [(Fall – 2006)]
MAK 206 Strength of Materials [(Fall – 2006) (Summer – 2008) (Spring – 2013, 2015, 2017)]
MAK 312 Machine Elements [(Fall - 2008, 2014, 2015) (Spring – 2010, 2012)]
MAK 404 Mechanical System Design [(Spring – 2009) (Fall – 2009, 2011) (Summer – 2012)]
MAK 444 Automotive Engineering [(Spring – 2013) (Summer – 2016)]
MAK 410 Finite Element Methods [(Fall – 2008)]
MAK 501 Engineering Mathematics [(Fall – 2010, 2011, 2012, 2014, 2015, 2016)]
MAK 506 Theory of Elasticity [(Spring – 2009) (Fall – 2010)]
MAK 510 Finite Element Methods [(Spring 2007) (Fall 2016)]

Near East University, **Nicosia, N. Cyprus**
ME 303 Machine Design I [(Fall – 2014, 2015, 2016)]
ME 304 Machine Design II [(Spring – 2015, 2016, 2017)]

American University of the Middle East (AUM), **Egaila, Kuwait**
ME 270 Basic Mechanics I [(Fall – 2022, 2023) (Spring– 2020, 2021, 2023)]
ME 309 Fluid Mechanics [(Fall – 2024)]
ME 323 Mechanics of Materials [(Fall – 2018, 2020) (Spring– 2020) (Summer– 2019)]
ME 352 Machine Design I [(Fall – 2018, 2019) (Spring – 2019, 2026)]
ME 375 System Modeling and Analysis [(Spring – 2024, 2026) (Fall - 2025) (Summer - 2026)]
ME 452 Machine Design II [(Fall – 2018, 2019, 2020, 2021, 2022,2023,2024) (Spring – 2019, 2020, 2021, 2022, 2023, 2024) (Summer– 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026)]
ME 463 Engineering Design [(Fall – 2020, 2022) (Spring – 2020, 2021, 2022, 2023, 2024, 2026)]
ME 200 Thermodynamics [Fall - 2025)]
PHYS 173 Modern Mechanics [Fall - 2025)]
CVL 297 Basic Mechanics: Statics [Fall - 2024, 2025)]
CVL 340 Hydraulics [Summer - 2025)]
CVL 231 Engineering Materials I [Spring - 2026]

Honors and Awards

Listed in Top 2% Scientists Worldwide by Stanford University, since 2020

Editorial Board Member, Thin-walled Structures, ELSEVIER, since 2018

Associate Editor, Measurement and Control, SAGE, since 2024

Editorial Review Board, Journal of Applied Biomaterials & Functional Materials, SAGE, since 2025

Fulbright Visiting Research scholarship to University of Arizona, 2013

The most cited articles for the paper entitled "The effect of geometrical parameters on the energy absorption characteristics of thin-walled structures under axial impact loading" in International Journal of Crashworthiness

The most cited articles published for the paper entitled "Multi-objective crashworthiness optimization of tapered thin-walled tubes with axisymmetric indentations" in Thin-Walled Structures

The most cited articles published from 2004 to 2008 for the paper entitled "Contact mechanics of graded coatings" in International Journal of Solids and Structures.

Granted Research Projects

E. Acar and M.A. Guler (2019-2021), "Design and optimization of a new energy efficient cylindrical roller bearing using contact mechanics and optimization techniques", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 35,000

M.A. Guler and D. Coker (2017 - 2020), "Numerical and experimental investigation of fracture propagation in composites with impact damage", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 100,000.

B. Yıldırım and M. A. Guler (2017 - 2018), "Design and optimization of the body and chasis of a new tram vehicle under dynamic and crash loads using Finite Element Method", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 50,000

M.A. Guler (2015 - 2016), "Examination of impact energy absorbing capacity of empty and aluminum based foam-filled crash boxes", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 10,000.

M.A. Guler (2013 - 2015), "Optimization of the structural parts and packaging of a dishwasher using Finite Element Method", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 135,000.

M.A. Guler and B. Aksoylu, (2013 - 2015), "Development of a method to predict strength and failure of layered composites using peridynamic theory". Budget $ 125,000.

M.A. Guler (2013), "Dynamic Analysis of a VSP Support Bracket as a senior design project", funded by OYAK RENAULT, Budget $ 1,500.

M.A. Guler (2013), "Design of a Shock Absorbing Component for a Front Bumper of an Automobile as a senior design project", funded by OYAK RENAULT, Budget $ 1,500.

M.A. Guler (2012), "Frontal crash analysis for the design of an automobile cockpit carrier as a senior design project", funded by OYAK RENAULT, Budget $ 1,500.

M.A. Guler, R.M. Görgülüarslan and Fırat Özer, (2011), "Side crash simulation analysis for monolithic production of reinforced parts as a senior design project", funded by OYAK RENAULT, Budget $ 3,000.

M.A. Guler and E. Acar, (2009 – 2011), "Accurate prediction and robust optimization of springback in dual phase steels during sheet metal forming operations", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 65,000.

M.A. Guler, (2008 – 2010), "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses", funded by Ministry of Industry and Trade of Turkey under the scope of SAN-TEZ project. Budget $ 225,000.

M.A. Guler and S. Dag, (2007 – 2009), "Computational and Analytical methods for contact mechanics analysis of functionally graded material", funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $ 70,000.

Professional Experience

The Scientific and Technological Council of Turkey (TÜBİTAK), **Ankara, Turkey**
Executive Committee Member for MAKİTEG Group **Apr, 2017 - Sep, 2018**

- Served as a committee member for evaluating research projects received from manufacturing industry submitted to Machinery, Manufacturing Technologies Group (MAKİTEG).

Executive Committee Member for USETEG Group **Apr, 2015 - Apr, 2016**

- Served as a committee member for evaluating research projects received from automotive industry submitted to for Transportation, Defense, Energy and Textile Technologies Group (USETEG).

Arçelik Dishwasher Inc., **Ankara, Turkey**
Project Consultant **Sep, 2013 - Sep, 2015**

- Principal investigator of the industrial project titled "Optimization of the structural parts and packaging of a dishwasher using Finite Element Method" funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK TEYDEB Project number 5150016).

Oyak Renault Inc., **Bursa, Turkey**
Project Consultant **Sep, 2011 - Jun, 2013**

- Helped Renault Engineers in doing FEA analysis of side crush analysis for the project titled "Renault Fluence mid-floor part monolithic design and production of the prototype by hot forming technology" funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK TEYDEB Project number 3110357)
- performed dynamic Analysis of a VSP support bracket
- designed a shock absorbing component for a front bumper
- conducted frontal crash analysis for the design of an automobile cockpit carrier

GE Marmara Technology Center (GE MTC) Inc., **Gebze, Turkey**
Consultant **Jan, 2010 - Apr, 2010**

- Helped GE MTC Engineers in doing FEA analysis of Aircraft Engine components using ANSYS

TEMSA GLOBAL Inc., **Gebze, Turkey**
Project Consultant **Sep, 2008 - Sep, 2010**

- Principal Investigator of the industrial project titled "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses" funded by Ministry of Industry and Trade of Turkey under the scope of SAN-TEZ project

TEMSA GLOBAL Inc., **Adana, Turkey**
Senior Structural Engineer **Mar, 2005 - Sep, 2006**

- Managed all phases of bus rollover simulation projects (ECE R66), including building the FEA models in ANSA, preparation of mass lists, verification of all necessary analysis data (e.g. center of gravity and mass moment of inertias of rigid components, engine, axles, fuel pump, AC etc), analysis setup in LS-DYNA, running, verification, and certification.
- Performed 3D stress analysis of automotive parts under structural and dynamic loads using ANSYS and determined the critical locations and suggested design changes to obtain maximum life of components.
- Carried out modal analysis of various type of busses and created animations to illustrate several modes of excitation.

CD-ADAPCO Inc. (Now SIEMENS), **Melville, NY, US**
Senior Structural and Thermal Engineer **Sep, 2000 - Sep, 2004**

- Reviewed contracts with customer to determine analysis requirements and goals and best methods to achieve them.
- Identified possible causes of failure of a glass lining in a reactor.
- Conducted single cylinder thermal analysis of a six-cylinder diesel engine to understand the thermal behavior of the cylinder and the head.

- Performed 3D stress analysis of rotating parts in the gas turbines and compressors under thermal, structural and dynamic loads using ANSYS.
- Carried out modal analysis of a generator and created a movie to illustrate several modes of excitation.
- Performed thermal and structural analysis of EGR coolers used in truck engines, identified the regions susceptible to cracking and recommended design changes to obtain maximum life of the components.
- Models included thermal and structural loading, intermittent contact, friction and non-linear material properties.
- Created numerous programs and scripts to automate model generation, analysis set-up, and verification.

Publications

Journal Papers

E Radi and MA Güler. Pin–hole contact with curvature-induced couple tractions in couple-stress elasticity. *International Journal of Solids and Structures*, page 114121, 2026

E Radi and MA Güler. Effects of microstructure in pin-loaded hole contact with clearance. *International Journal of Engineering Science*, 221:104454, 2026

Özlem Akçay, Murat Altın, Mehmet Ali Güler, and Erdem Acar. Machine learning based crashworthiness optimization of 3d-printed hybrid multi-cell energy absorber. *European Journal of Mechanics-A/Solids*, page 106109, 2026

Fei Shen, You-Hua Li, Mehmet Ali Güler, Hong-Dong Wu, Wei-Wei Shen, and Liao-Liang Ke. A high-efficiency prediction method for thermal contact resistance of rough surfaces. *International Communications in Heat and Mass Transfer*, 167:109325, 2025

You-Hua Li, Liao-Liang Ke, Gang-Gang Chang, Mehmet Ali Güler, and Fei Shen. On the electrical contact resistance model of rough surfaces based on multi-physics modeling. *International Journal of Solids and Structures*, page 113583, 2025

You-Hua Li, Liao-Liang Ke, Kun Zhou, Gang-Gang Chang, Mehmet Ali Güler, Wei-Wei Shen, and Fei Shen. On the multi-physics elastoplastic electrical contact of rough surfaces. *Tribology International*, 204:110418, 2025

Ali Mamedov, Ali Dinc, Mehmet Ali Guler, Murat Demiral, and Murat Otkur. Tool wear in micro-milling: a review. *The International Journal of Advanced Manufacturing Technology*, 137(1):47–65, 2025

Lovejoy Mutswatiwa, Mehmet Ali Güler, and Celestin Nkundineza. An investigation on the impact of non-uniform track stiffness on wheel-rail interaction mechanics. *Journal of Vibration Engineering & Technologies*, 13(7):518, 2025

Ahmed Saber, Mehmet Ali Güler, Erdem Acar, Omar Soliman ElSayed, Hussain Aldallal, Abdulrahman Alsadi, and Yousef Aldousari. Crash performance of additively manufactured tapered tube crash boxes: influence of material and geometric parameters. *Designs*, 9(3):72, 2025

Oguzhan Mulkoglu, Mehmet Ali Güler, and Fatih Göncü. Modified hot-spot stress method for fatigue life estimation of welded components. *Materials Testing*, 67(5):821–832, 2025

Mehmet Ali Güler and Yadolah Alinia. An elastic-constant stress cohesive zone model for an orthotropic piezoelectric actuator adhesively bonded to a substrate. *Engineering Fracture Mechanics*, page 110391, 2024

Ahmed Saber, Mehmet Ali Güler, Murat Altin, and Erdem Acar. Bio-inspired thin-walled energy

absorber adapted from the xylem structure for enhanced vehicle safety. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46(10):613, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. An efficient method for electro-thermo-mechanical coupling effect in electrical contact on rough surfaces. *International Journal of Heat and Mass Transfer*, 226:125492, 2024

Dengke Li, Peijian Chen, Hao Liu, Zhilong Peng, Mehmet Ali Guler, and Shaohua Chen. The interfacial behavior of an axisymmetric film bonded to a graded inhomogeneous substrate. *Mechanics of Materials*, page 104983, 2024

Erdem Acar, Evren Emre Üstün, Mehmet Ali Güler, Ahmet Toros, and Ozan Müştak. Design of an energy efficient cylindrical roller bearing. *Mechanics Based Design of Structures and Machines*, 52(2):826–847, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. Modeling multi-physics electrical contact on rough surfaces considering elastic-plastic deformation. *International Journal of Mechanical Sciences*, page 109066, 2024

Mehmet Ali Güler, Yadolah Alinia, and Enrico Radi. Couple stress effects in a thin film bonded to a half-space. *Mathematics and Mechanics of Solids*, 29(4):629–644, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. A rough surface electrical contact model considering the interaction between asperities. *Tribology International*, 190:109044, 2023

Mehmet Ali Güler, Sinem K Mert, Murat Altin, and Erdem Acar. An investigation on the energy absorption capability of aluminum foam-filled multi-cell tubes. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 45(10):541, 2023

Mehmet Ali Güler, Yadolah Alinia, and Ergun Nart. Interfacial delamination for an orthotropic thin film/substrate system. *Theoretical and Applied Fracture Mechanics*, page 104007, 2023

Seyed Ali Abbaszadeh-Fathabadi, Yadolah Alinia, and Mehmet Ali Güler. On the mechanics of a double thin film on a finite thickness substrate. *International Journal of Solids and Structures*, page 112349, 2023

E Radi, A Nobili, and MA Guler. Indentation of a free beam resting on an elastic substrate with an internal lengthscale. *European Journal of Mechanics-A/Solids*, page 104804, 2022

Sinem K Mert, Mehmet Ali Güler, Murat Altin, Erdem Acar, and Adem Çiçek. Crashworthiness evaluation of hybrid tubes made of gfrp composite and aluminum tubes filled with honeycomb structures. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 45(6):291, 2023

Samet Eltaş, Mehmet Ali Güler, Konstantinos Daniel Tsavdaridis, Christos E Sofias, and Bora Yıldırım. On the beam-to-beam eccentric end plate connections: A numerical study. *Thin-Walled Structures*, 188:110787, 2023

Qing-Hui Luo, Yue-Ting Zhou, and Mehmet Ali Guler. Adhesive behavior between dissimilar materials subjected to thermo-elastic loadings with normal-tangential coupling effect. *Applied Mathematical Modelling*, 115:360–384, 2023

Ugur Yolum, Mirac Onur Bozkurt, Eda Gok, Demirkan Coker, and Mehmet Ali Güler. Crack propagation in the double cantilever beam using peridynamic theory. *Composite Structures*, page 116050, 2022

Oguzhan Mulkoglu, Alireza Seyedi, Mehmet Ali Güler, Bora Yildirim, and Fatih Göncü. Fatigue life estimation of welded connections in a tram vehicle. *International Journal of Heavy Vehicle Systems*, 29(5):537–552, 2022

Cüneyt Aktaş, Erdem Acar, Mehmet Ali Güler, and Murat Altın. An investigation of the crashworthiness performance and optimization of tetra-chiral and reentrant crash boxes. *Mechanics Based Design of Structures and Machines*, pages 1–24, 2022

Yadolah Alinia and Mehmet Ali Güler. On the problem of an axisymmetric thin film bonded to a transversely isotropic substrate. *International Journal of Solids and Structures*, 248:111636, 2022

İsa Çömez and Mehmet Ali Güler. Thermoelastic contact problem of a rigid punch sliding on a functionally graded piezoelectric layered half plane with heat generation. *Journal of Thermal Stresses*, 45(3):191–213, 2022

I Çömez, Y Alinia, and MA Güler. The partial slip contact problem for a monoclinic coated half plane. *Archives of Mechanics*, 74(1):13–37, 2022

Sami El-Borgi, Isa Çömez, and Mehmet Ali Güler. A receding contact problem between a graded piezoelectric layer and a piezoelectric substrate. *Archive of Applied Mechanics*, 91(12):4835–4854, 2021

E Nart, Y Alinia, and MA Güler. The effect of material anisotropy on the mechanics of a thin-film/substrate system under mechanical and thermal loads. *Mathematics and Mechanics of Solids*, page 10812865211031277, 2021

M Kathiresan, K Manisekar, Vasudevan Rajamohan, and Mehmet A Güler. Investigations on crush behavior and energy absorption characteristics of gfrp composite conical frusta with a cutout under axial compression loading. *Mechanics of Advanced Materials and Structures*, pages 1–18, 2021

Barış Mehmet Zeytinci, Mehmet Şahin, Mehmet Ali Güler, and Konstantinos Daniel Tsavdaridis. A practical design formulation for perforated beams with openings strengthened with ring type stiffeners subject to vierendeel actions. *Journal of Building Engineering*, page 102915, 2021

Sabri Alper Keskin, Erdem Acar, MA Güler, and Murat Altin. Exploring various options for improving crashworthiness performance of rail vehicle crash absorbers with diaphragms. *Structural and Multidisciplinary Optimization*, pages 1–16, 2021

E Özaslan, A Yetgin, B Acar, and MA Güler. Damage mode identification of open hole composite laminates based on acoustic emission and digital image correlation methods. *Composite Structures*, page 114299, 2021

İ Çömez, Yadollah Alinia, MA Güler, and Sami El-Borgi. Partial slip contact analysis for a monoclinic half plane. *Mathematics and Mechanics of Solids*, 26(3):401–421, 2021

Murat Altin, Erdem Acar, and MA Güler. Crashworthiness optimization of hierarchical hexagonal honeycombs under out-of-plane impact. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, 235(6):963–974, 2021

Ugur Yolum, Demirkan Coker, and Mehmet A Güler. Intersonic shear crack propagation using peridynamic theory. *International Journal of Fracture*, 228(1):103–126, 2021

Sinem K Mert, Murat Demiral, Murat Altin, Erdem Acar, and Mehmet A Güler. Experimental and numerical investigation on the crashworthiness optimization of thin-walled aluminum tubes considering damage criteria. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 43(2):1–22, 2021

Y Alinia, SA Abbaszadeh-Fathabadi, and MA Guler. Stress analysis for a thin film bonded to an orthotropic substrate under thermal loading. *Mechanics Research Communications*, 109:103594, 2020

Erdem Acar, Burak Yilmaz, Mehmet A Güler, and Murat Altin. Multi-fidelity crashworthiness optimization of a bus bumper system under frontal impact. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 42(9):1–17, 2020

Mehmet Ali Güler, Muhammed Emin Cerit, Sinem Kocaoglan Mert, and Erdem Acar. Experimental and numerical study on the crashworthiness evaluation of an intercity coach under frontal impact conditions. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, page 0954407020927644, 2020

I Çömez and MA Güler. On the contact problem of a moving rigid cylindrical punch sliding over an orthotropic layer bonded to an isotropic half plane. *Mathematics and Mechanics of Solids*, page 1081286520915272, 2020

İsa Çömez, Mehmet Ali Güler, and Sami El-Borgi. Continuous and discontinuous contact problems of a homogeneous piezoelectric layer pressed by a conducting rigid flat punch. *Acta Mechanica*, 231(3):957–976, 2020

Ender İnce and Mehmet A Güler. On the advantages of the new power-split infinitely variable transmission over conventional mechanical transmissions based on fuel consumption analysis. *Journal of Cleaner Production*, 244:118795, 2020

Ugur Yolum and MA Güler. On the peridynamic formulation for an orthotropic mindlin plate under bending. *Mathematics and Mechanics of Solids*, 25(2):263–287, 2020

Mürvet Bektaş, Mehmet Ali Güler, and Dilek Funda Kurtuluş. One-way fsi analysis of bio-inspired flapping wings. *International Journal of Sustainable Aviation*, 6(3):172–194, 2020

B Yildirim, KB Yilmaz, I Comez, and MA Guler. Double frictional receding contact problem for an orthotropic layer loaded by normal and tangential forces. *Meccanica*, 54(14):2183–2206, 2019

K.B. Yilmaz, İ Çömez, M.A. Güler, and B. Yildirim. Sliding frictional contact analysis of a monoclinic coating/isotropic substrate system. *Mechanics of Materials*, 137:103132, 2019

E. Acar, M. Altin, and M.A. Güler. Evaluation of various multi-cell design concepts for crashworthiness design of thin-walled aluminum tubes. *Thin-Walled Structures*, 142:227–235, 2019

E. Özaslan, M.A. Güler, A. Yetgin, and B. Acar. Stress analysis and strength prediction of composite laminates with two interacting holes. *Composite Structures*, 221:110869, 2019

I. Comez, K.B. Yilmaz, M.A. Güler, and B. Yildirim. On the plane frictional contact problem of a homogeneous orthotropic layer loaded by a rigid cylindrical stamp. *Archive of Applied Mechanics*, pages 1–17, 2019

K.B. Yilmaz, I. Comez, M.A. Güler, and B. Yildirim. The effect of orthotropic material gradation on the plane sliding frictional contact mechanics problem. *The Journal of Strain Analysis for Engineering Design*, 54(4):254–275, 2019

E. Acar, N. Tüten, and M.A. Güler. Design optimization of an automobile torque arm using global and successive surrogate modeling approaches. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, 233(6):1453–1465, 2019

E. İnce and M.A. Güler. Design and analysis of a novel power-split infinitely variable power transmission system. *Journal of Mechanical Design*, 141(5):054501, 2019

M. Altin, Ü. Kılınçkaya, E. Acar, and M.A. Güler. Investigation of combined effects of cross section, taper angle and cell structure on crashworthiness of multi-cell thin-walled tubes. *International Journal of Crashworthiness*, 24(2):121–136, 2019

K.B. Yilmaz, İ. Çömez, M.A. Güler, and B. Yildirim. Analytical and finite element solution of the sliding frictional contact problem for a homogeneous orthotropic coating-isotropic substrate system. *ZAMM-Journal of Applied Mathematics and Mechanics/Zeitschrift für Angewandte Mathematik und Mechanik*, 99(3):e201800117, 2019

M. Altin, E. Acar, and M.A. Güler. Foam filling options for crashworthiness optimization of thin-walled multi-tubular circular columns. *Thin-Walled Structures*, 131:309–323, 2018

Y. Alinia, M. Hosseini-nasab, and M.A. Güler. The sliding contact problem for an orthotropic coating bonded to an isotropic substrate. *European Journal of Mechanics-A/Solids*, 70:156–171, 2018

K.B. Yilmaz, I. Comez, B. Yildirim, M.A. Güler, and S. El-Borgi. Frictional receding contact problem for a graded bilayer system indented by a rigid punch. *International Journal of Mechanical Sciences*, 141:127–142, 2018

Y. Alinia and M.A. Güler. On the fully coupled partial slip contact problems of orthotropic materials loaded by flat and cylindrical indenters. *Mechanics of Materials*, 114:119 – 133, 2017

H. Zakerhaghighi, S. Adibnazari, M.A. Güler, and S.A. Faghidian. Two-dimensional analysis of the fully coupled rolling contact problem between a rigid cylinder and an orthotropic medium. *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik*, 97(10):1283–1304, 2017

M. Altin, M.A. Güler, and S.K. Mert. The effect of percent foam fill ratio on the energy absorption capacity of axially compressed thin-walled multi-cell square and circular tubes. *International Journal of Mechanical Sciences*, 131-132:368 – 379, 2017

A.E. Orun and M.A. Guler. Effect of hole reinforcement on the buckling behaviour of thin-walled beams subjected to combined loading. *Thin-Walled Structures*, 118:12 – 22, 2017

I. Comez and M.A. Guler. The contact problem of a rigid punch sliding over a functionally graded bilayer. *Acta Mechanica*, 228(6):2237–2249, 2017

O. Mülkoğlu, M.A. Güler, E. Acar, and H. Demirbağ. Drop test simulation and surrogate-based optimization of a dishwasher mechanical structure and its packaging module. *Structural and Multidisciplinary Optimization*, 55(4):1517–1534, 2017

Y. Alinia, H. Zakerhaghighi, S. Adibnazari, and M.A. Güler. Rolling contact problem for an orthotropic medium. *Acta Mechanica*, 228(2):447–464, Feb 2017

M.A. Güler, A. Kucuksucu, K.B. Yilmaz, and B. Yildirim. On the analytical and finite element solution of plane contact problem of a rigid cylindrical punch sliding over a functionally graded orthotropic medium. *International Journal of Mechanical Sciences*, 120:12 – 29, 2017

A Taştan, Erdem Acar, Mehmet Ali Güler, and Ü Kılınçkaya. Optimum crashworthiness design of tapered thin-walled tubes with lateral circular cutouts. *Thin-Walled Structures*, 107:543–553, 2016

R. Elloumi, S. El-Borgi, M.A. Guler, and I. Kallel-Kamoun. The contact problem of a rigid stamp with friction on a functionally graded magneto-electro-elastic half-plane. *Acta Mechanica*, 227(4):1123, 2016

Y. Alinia, A. Beheshti, M.A. Guler, S. El-Borgi, and A. Polycarpou. Sliding contact analysis of functionally graded coating/substrate system. *Mechanics of Materials*, 94:142–155, 2016

A. Kucuksucu, M.A. Guler, and A. Avci. Mechanics of sliding frictional contact for a graded orthotropic half-plane. *Acta Mechanica*, 226(10):3333, 2015

S. El-Borgi, S. Usman, and M.A. Güler. A frictional receding contact plane problem between a functionally graded layer and a homogeneous substrate. *International Journal of Solids and Structures*, 51(25):4462–4476, 2014

M.A. Guler. Closed-form solution of the two-dimensional sliding frictional contact problem for an orthotropic medium. *International Journal of Mechanical Sciences*, 87:72 – 88, 2014

R. Elloumi, I. Kallel-Kamoun, S. El-Borgi, and M.A. Guler. On the frictional sliding contact problem between a rigid circular conducting punch and a magneto-electro-elastic half-plane. *International Journal of Mechanical Sciences*, 87:1 – 17, 2014

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A.T. Camcı, M.A. Guler, (2014), "Hafif Elektrikli Araçlarda Sürekli Değişken Oranli Şanziman Kullanilarak Menzil Ve Performansin Artirilmasi", 7th Automotive Technologies Congress, Bursa, Turkey.
O. Üşenmez, M. Tekir, Y.Z. Akman, N. Sezer-Uzol, M.A. Güler, (2012), "Sessiz Helikopter Tasarim Ve Analizi Için Izlenen Adimlar Ve Yöntemler, SAVTEK 2012, 6. Savunma Teknolojileri Kongresi, Ankara, Turkey.

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K. Elitok, M.A. Guler, F.H. Avcı and U. Stelzmann, (2005), "Regulatory Bus Roll-Over Crash Analysis Using LS-DYNA", Conference for Computer-Aided Engineering and System Modeling, İstanbul, Turkey.

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E. Radi, M.A. Guler, (2024), "Effects of couple tractions on contact problems at the microscale", AIMETA Conference, Napoli, Italy.

U. Yolum, M.A. Guler, (2017), "A Peridynamic Plate Formulation to Couple Delamination and in-plane Failures for Composite Laminates", AFM 2017 International Conference on Advances in functional Materials, Los Angeles, USA.

D.J. Bang, M.A. Guler, E. Madenci, (2014), "Ordinary state-based peridynamic modeling of hyperelastic materials", ASME 2014 International Mechanical Engineering Congress and Exposition, Montreal, Canada.

S. Adibnazari, M.A. Guler, Y. Alinia, (2011), "Rolling Contact Mechanics of FGM Coatings", ASME 2011 International Mechanical Engineering Congress and Exposition, Denver, Colorado, USA.

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M.A. Guler, F. Ozer, M. Yenice and M. Kaya, (2010), "Springback Prediction of DP600 Steels for Varius Material Models, The 13th International Conference on Metal Forming, Toyahashi, Japan.

T. Apatay, S. Dag, M.A. Guler and M. Gulgec, (2010), "Subsurface stresses in an FGM coating loaded by a sliding flat punch", Fourth European Conference on Computational Mechanics, Minisymposium on Fracture and Contact Mechanics for Interface Problems, Paris, France.

M.A. Guler, (2008), "Mechanical Failure of Thin Films Bonded to Graded Substrates", The Mechanics Conference to Celebrate the 100th Anniversary of the Department of Engineering Science and Mechanics, Professor Liviu Librescu Memorial Sessions, Virginia Tech, Blacksburg, Virginia, USA.

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F. Erdogan, S. Dag and M.A. Guler, (2000), "Contact and crack problems in functionally graded materials", 20th International Congress of Theoretical and Applied Mechanics, August 2000, Chicago, Illinois, USA.

Post Doc. Researcher

A. Kucuksucu, "Contact mechanics of orthotropically graded materials", 2012-2014.

Visiting Research Scholar

Y. Alinia, "Rolling contact mechanics of graded materials", 2011-2012.

Supervised Ph.D. Students

S.K. Mert, (2022) "Experimental and Numerical Investigation of Energy Absorbing Capacity of Aluminum, Composite, and Hybrid Crash Boxes", (Co-supervised with Dr. Adem Çiçek)

O. Mulkoglu, (2022) "Fatigue Life Estimation of Welded Structures: A New Modified Hot-spot Stress Method", (Co-supervised with Dr. Fatih Göncü)

U. Yolum, (2021), "Investigation of bending and mod II failures in thick plates using peridynamic theory"

E. Ince, (2019), "Design of a new power-split infinitely variable transmission"

M. Altın, (2017), "Investigation of the energy absoption capacities of metallic foam filled crash-boxes used in automobiles". (Co-supervised with Dr. H.S. Yücesu)

Supervised M.Sc. Students

S. Eltaş, (2023) "Finite Element Analysis of Beam-to-Beam Eccentric End Plate Connections", (Co-supervised with Dr. Bora Yıldırım)

S.A. Keskin (2021) "Studying and optimizing the effeccts of different design options on the performance of crash absorbers with diagrams", (Co-supervised with Dr. Erdem Acar)

B.B. Mercan (2021) "Investigation of holed composite plates exposed to in-plane loading by the peridynamic theory", (Co-supervised with Dr. Bora Yıldırım)

E. Gök (2020) "Investigation of damage behavior in composite materials under mode I and mode II loads using peridynamic theory"

M. Bektaş (2020) "Aeroelastic analysis of bio-inspired flapping wing models", (Co-supervised with Dr. D. F. Kurtuluş)

B. Yılmaz (2019) "Application of multi-fidelity modeling technique on crash analyses", (Co-supervised with Dr. E. Acar)

B.M. Zeytinci (2019) “Investigation of the effect of the ring type stiffeners on the vierendeel mechanism of perforated beams under bending”

E. Özaslan (2019) “Investigation of mechanical strength of open hole composite plates with finite element method"

A.R. Seyedi (2019) "Design and optimization for body and chassis of new tram with finite element method under static, dynamic and collision loads.", (Co-supervised with Dr. B. Yıldırım)

F. Can (2018), “The design of Flow controller unit for Joule-Thomson type crycooler”, (Co-supervised with Dr. M. K. Aktaş)

A.E. Örün (2017), “Effect of hole reinforcement on buckling behavior under combined loads for a beam structure”

K.B. Yılmaz (2016), “Finite element analysis of contact mechanics of rigid punches sliding over a graded orthotropic medium”, (Co-supervised with Dr. B. Yıldırım)

N. Tüten (2016), “Weight optimization and reliability prediction of an automobile torque arm subjected to cyclic loading”, (Co-supervised with Dr. E. Acar)

A Tastan (2016), “Development of a method to predict strength and failure of isotropic and composite structures under tension and bending loadings using bond-based peridynamic theory”

O Mülkoğlu (2016), “Optimization of the mechanical structure of a dishwasher and its packaging module using Finite Elment Method”

T Kiper (2015), “Analysis of bird strike effects on the wing of a training aircraft”, (Co-supervised with Dr. İ. Uslan)

O.M. Bircan (2014), “An analytical approach to design form-turning insert profiles: Investigation of rake face design on tool life”, (Co-supervised with Dr. Y. Karpat)

A.T. Camcı (2013), “Improvements on the range and the performance of a light electric vehicle by making use of a continuously variable transmission”

F. Özer, (2011), “Springback prediction in Advanced High Strength Steels after forming operations”, (Co-supervised with Dr. İ. B. Özsoy)

D. Bekar, (2011), “Robust optimization of springback predictions in dual phase steels during sheet metal forming operations”, (Co-supervised with Dr. E. Acar)

L. Sözen, (2011), “Springback prediction in rotary tube bending operations”

M.E. Cerit, (2011), "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses"

Y.F. Gülver, (2009), "Mechanical modeling of thin films bonded to functionally graded coatings".

External Ph.D. Committees

Cong Tien Nguyen, Naval Architecture, Ocean and Marine Engineering University of Strathclyde, Ph.D. thesis defence external comittee member, November 2020.
Title of the thesis: Peridynamics for damage prediction in ships and offshore structures

Tarek Salaheldin Shaban, Department of Mechanical Engineering, Qatar University, M.Sc. thesis defence external comittee member, April 2019.
Title of the thesis: Stress Shielding Reduction Approach for Femoral Stems used in Total Hib Arthroplasty (THA) using topology optimization.

Al-Sendibad Said Shehadeh, Department of Mechanical Engineering, Qatar University, M.Sc. thesis defence external comittee member, April 2019.
Title of the thesis: Evaluation of 316L Stainless Steel Part fabrication using additive and subtractive manufacturing: a guideline for process selection

Onur Arslan, Department of Mechanical Engineering, Middle East Technical University, Ph.D. thesis defence external committee member, June 2017.

Jabbar Gattmah, Department of Mechanical Engineering, Yıldırım Beyazıt University, Ph.D. thesis defence external committee member, June 2017.

Community Service

Editorial Board Member, Thin-walled Structures, since 2018
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Reviewer, International Journal of Crashworthiness,
Reviewer, International Journal of Fatigue,
Reviewer, Engineering Fracture Mechanics
Reviewer, ZAMM
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Reviewer for industrial and academic projects for TUBITAK (2007 - present)

References

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